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Ho Chi Minh City, Vietnam

Education

2023–26¹ **B.Sc.**, Software Engineering, FPT University

Research Experience

2023–26² Research Assistant, [AI Technology and Application Research Lab](#), FPT University,
Ho Chi Minh City, Vietnam

Awards & Honors

Dec 2025 Second Prize, Student Research Festival Fall 2025, FPT University
Aug 2025 Second Prize, Student Research Festival Summer 2025, FPT University

Publications

 [Google Scholar](#)

† → Equal contribution

Journal Articles

- J1. **Duc Tai Phan**, Nguyen, N. M., Nguyen, K. P., Tran, P.-N., Pham, N. T., Le, L., Hong, C. S. & Dang, D. N. M. From object difficulty to image scoring: A strategy for active learning in object detection. *Knowledge-Based Systems*. <https://doi.org/10.1016/j.knosys.2026.115946> (2026).

Peer-reviewed Conference Proceedings

- C1. Nguyen, T. T., Nguyen, N. M., **Duc Tai Phan**, Hoang, Q. N., Pham, T. M. & Dang, D. N. M. *Swin Transformer V2 for Optical Chemical Structure Recognition: Comparison with Convolutional Neural Networks and Swin Transformer Variants in The 22nd International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON 2025)* (2026).

¹Expected.

- C2. Nguyen, P. L., **Duc Tai Phan**, Nguyen, T. T., Nguyen, N. M. & Dang, D. N. M. YOLOv5-Powered Smart Parking System with IoT-Based Real-Time Slot Monitoring in 2025 RIVF International Conference on Computing and Communication Technologies (RIVF 2025) (2025). <https://doi.org/10.1109/RIVF68649.2025.11365041>.
- C3. **Duc Tai Phan**, Nguyen, N. M. & Dang, D. N. M. DAAL: Dual Ambiguity in Active Learning for Object Detection with YOLOE in 17th International Conference on Management of Digital Ecosystems (2025).
- C4. Nguyen, N. M., **Duc Tai Phan** & Dang, D. N. M. GloMER: Towards Robust Multimodal Emotion Recognition via Gated Fusion and Contrastive Learning in 17th International Conference on Management of Digital Ecosystems (2025).
- C5. **Duc Tai Phan**, Nguyen, N. M., Nguyen, K. P., Pham, T. M. & Dang, D. N. M. ALMUS: Enhancing Active Learning for Object Detection with Metric-Based Uncertainty Sampling in 2025 25th Asia-Pacific Network Operations and Management Symposium (APNOMS'2025) (2025). <https://doi.org/10.23919/APNOMS67058.2025.11181447>.
- C6. Nguyen, N. M., Le, T. T., Nguyen, T. T., **Duc Tai Phan**, Tran, A. K. & Dang, D. N. M. Ce-moBAM: Advancing Multimodal Emotion Recognition through Heterogeneous Graph Networks and Cross-Modal Attention Mechanisms in 2025 25th Asia-Pacific Network Operations and Management Symposium (APNOMS'2025) (2025). <https://doi.org/10.23919/APNOMS67058.2025.11181320>.
- C7. **Duc Tai Phan**, Tran, P.-N. & Dang, D. N. M. Improving Face Attendance Checking System with Ensemble Learning in 2024 RIVF International Conference on Computing and Communication Technologies (RIVF 2024) (2024). <https://doi.org/10.1109/RIVF64335.2024.11009085>.
- C8. Tran, P.-N., Pham, N. T., Phan, N. V. H., **Duc Tai Phan**, Nguyen, C. T. & Dang, D. N. M. Towards Real-Time Vietnamese Traffic Sign Recognition on Embedded Systems in 2024 15th International Conference on Information and Communication Technology Convergence (ICTC 2024) (2024). <https://doi.org/10.1109/ICTC62082.2024.10827558>.

Industry Experience

2025–26	AI Engineer, IMT Solutions , Ho Chi Minh City, Vietnam
2024–25	AI Engineer, ABCStudio , Ho Chi Minh City, Vietnam